

Observing Obfuscation In Multiple Different Angles

Ensemble Detection of Obfuscated Prompt Injection

Bashh Dazer

Flara Research Lab · dev@bashh.smail.me

Abstract

Obfuscated prompt injection induces a measurable structural signature: encoding-based payloads tend to fragment into multiple semantically distinct chunks at sentence boundaries, while benign text remains predominantly single-chunk (99% vs 28.3% multi-chunk rate). We test whether embedding geometry can exploit this signature, and how much value it adds beyond a trivial structural feature such as chunk count. We present a four-feature ensemble detector combining sentence- and paragraph-level embedding delta angles, an optimised regex coverage signal, and a single-chunk Unicode anomaly bonus, evaluated on N=1,152 samples across 21 obfuscation types (Section 4.1). The ensemble achieves AUC 0.974, F1 0.960, and 94.7% detection at 1% FPR on the full dataset (weights and threshold also fit on the full dataset); a 70/30 held-out split with weights fit on the training portion only reaches AUC 0.958 (Section 4.4), confirming the result generalises rather than overfits. Headline figures throughout this paper are the full-dataset numbers unless stated otherwise. The ensemble also achieves 100% detection of hex and backslash-visual encoding (Table 2).

Delta angle correlates substantially with chunk count and word count ($r=0.68$, $r=0.47$) and weakly with character-level features such as entropy ($r=-0.08$): a direct substitution test shows the embedding signal still beats raw chunk count, but by a modest, real margin (1.0-1.4pp AUC) that is larger for some embedding models than the production model (Section 5.4). Two of the four ensemble features do not mean what their formula weight implies: paragraph delta carries substantial weight despite a near-chance standalone AUC, because regex and the Unicode bonus already cover every multi-paragraph attack in this dataset; sentence delta carries a small weight yet is costly to remove (Section 5.2). The Unicode anomaly signal addresses 46.9% of the single-chunk blind spot at zero false positives. A re-optimised character entropy baseline is a stronger competitor than previously reported (AUC 0.722) but shares regex's blind spot on the multi-chunk attack types delta angle catches.

An adaptive-attack evaluation (30 representative samples, 20 evasion strategies) finds that plain natural-language prompt injection evades the detector entirely -- 20/30 (66.7%) at 1% FPR -- with catches concentrated almost exclusively in Unicode/homoglyph and structural tricks. This sharpens the threat model: the regex keyword list targets encoding vocabulary (decode, translate, payload), not injection phrasing (ignore, override, system prompt), so social-engineering injection is out of scope for this detector and requires a separate intent-detection layer. Embedding geometry therefore provides a modest but real advantage over chunk-counting for detecting structural obfuscation specifically; it is not a general prompt-injection detector (Appendix C), and is designed as a Stage 1 preprocessing filter in a defence-in-depth architecture, not a standalone classifier.

1. Introduction

Obfuscated prompt injections -- attacks that encode their payload through character encoding (hex, base64, backslash escape sequences) or structural techniques (dot-spacing, Unicode substitution) to evade detection -- often induce a measurable multi-segment structural signature that benign text does not share. We investigate whether embedding geometry can exploit this signature, and whether doing so adds detection value beyond a trivial structural feature such as chunk count. As language models are increasingly deployed as autonomous agents with access to tools, memory, and external services, detecting this class of injection before it reaches the model is a practical, narrow problem worth solving precisely, rather than folding into a general prompt-injection classifier.

Pattern-based regex handles encoding-based attacks reasonably but has no mechanism for any technique that produces no distinctive character-level signature -- it cannot see chunk structure. We introduce a complementary signal, the **delta angle** between consecutive embedding chunks of the input text: encoding-based payloads generate multiple semantically distinct chunks with measurable angular separation, while benign inputs are predominantly single-chunk (99% vs 28.3% multi-chunk rate). This work grew out of an earlier, more general attempt to use delta angle as a standalone anti-injection measure, which did not separate normal from injected prompts reliably; Appendix C documents that prior work and the reasoning behind narrowing the scope to obfuscation detection specifically.

Our main findings are:

- The structural signature is real and exploitable: a four-feature ensemble (sentence- and paragraph-level delta angle, optimised regex, single-chunk Unicode anomaly bonus) reaches AUC 0.974, F1 0.960, 94.7% detection at 1% FPR (Section 5.2).
- Delta angle correlates substantially with chunk and word count ($r=0.68$, $r=0.47$); the embedding signal beats a trivial chunk-counter, but by a modest, directly quantified margin rather than full independence from input structure -- and the margin is model-dependent (Section 5.3-5.4).
- Ensemble weight magnitude is not a reliable proxy for a feature's marginal contribution: two of the four features (paragraph delta, sentence delta) each contradict what their formula weight would suggest, in opposite directions (Section 5.2).
- The Unicode anomaly signal addresses 46.9% of the single-chunk blind spot at zero false positives (Section 6.2); eleven further geometric extensions were tested and did not improve on the production ensemble (Appendix B).
- An adaptive red-team evaluation (30 representative attacks, 20 evasion strategies) shows the detector's blind spot is plain natural-language injection, not exotic encodings -- it is a Stage 1 structural filter, not a general prompt-injection classifier (Section 5.6, Section 7).

2. Related Work

Prompt injection as a named threat was introduced by Perez and Ribeiro (2022), with subsequent taxonomy work by Greshake et al. (2023) distinguishing direct and indirect injection. The HackAPrompt dataset (Schulhoff et al., 2023) provides a benchmark for injection resistance, though it does not focus specifically on encoding-based obfuscation.

Existing detection approaches fall into two categories. *Input-side detectors* analyse user input before it reaches the model, including perplexity-based filters (Alon and Kamfonas, 2023), instruction-following probes, and classifier-based approaches. *Output-side detectors* analyse model behaviour for signs of successful injection. Our approach is an input-side detector operating on structural properties of the input text.

The use of embedding geometry for text analysis has precedent in semantic similarity, anomaly detection, and out-of-distribution detection. To our knowledge, applying consecutive embedding chunk angles as an obfuscation signal is novel. The closest related work uses perplexity or entropy as input-side signals; we show that delta angle captures independent signal not available from these measures.

3. Method

3.1 Sentence Delta Angle

Given an input text x , we split it into chunks using sentence boundaries (regex: $(?<=[.!?])\s+$), followed by clause boundaries for sentences exceeding 50 characters. Chunks below 8 words are merged with their neighbour. The merge threshold of 8 words was selected via distributional analysis: the chunk length distribution shows a natural density trough at the 7 to 8 word boundary, with 40.8% of comma-split chunks falling at exactly 7 words (Figure 3).

Each chunk is embedded via the NVIDIA NIM API using `nvidia/nv-embedqa-e5-v5` (4096 dimensions). The unsigned angle between consecutive chunk embeddings is:

$$\text{theta}(i) = \arccos(\text{dot}(\mathbf{e}_i, \mathbf{e}_{(i+1)}) / (||\mathbf{e}_i|| * ||\mathbf{e}_{(i+1)}||))$$

The sentence delta is the mean of all consecutive pair angles, excluding zero-angle pairs. Unsigned angles are used throughout; Appendix B.3 confirms that signed angles produce model-dependent noise.

3.2 Paragraph Delta Angle

Paragraph delta is computed identically but uses paragraph boundaries (blank lines) as chunk delimiters, with no further word-count merging. This produces coarser, sparser chunking than the sentence-level signal: most inputs contain no blank-line break at all, so paragraph delta is defined (non-zero) for only 22/1,152 samples in our dataset (Section 5.2), versus the majority-multi-chunk rate of sentence delta on obfuscated text (Section 4.1). Across the full dataset, sentence and paragraph delta correlate at 0.39, but this is driven mostly by which samples are multi-chunk at all; restricted to only the 22 samples where paragraph delta is actually non-zero, the correlation rises to 0.64 (n too small to treat as precise, but the two signals are evidently related, not independent, where paragraph delta applies). Whether paragraph delta adds anything beyond what sentence delta and the other features already capture is an empirical question, addressed in Section 5.2.

3.3 Regex Coverage Signal

We define 9 regex patterns targeting encoding artifacts and instruction-keyword density. Each pattern produces a 0-1 float score proportional to matched character coverage. The final regex score is a weighted sum with weights optimised via differential evolution targeting detection rate at 1% FPR (Table 1).

Pattern	Target	Weight
<code>special_chars</code>	Non-alphanumeric character ratio	0.179
<code>digit_escape</code>	Digit escapes (<code>\32</code> , <code>\97</code>)	0.159
<code>backslash_density</code>	Backslash density (count / total chars)	0.146
<code>repetition</code>	Line repetition ratio	0.145
<code>hex</code>	Hex escapes (<code>\x41\x42</code>)	0.125
<code>consecutive_backslash</code>	Consecutive backslash chains	0.090
<code>keywords</code>	29 attack-instruction terms plus 6 connective patterns	0.079
<code>backslash</code>	Backslash-visual encoding (<code>\ \h \a \v \e</code>)	0.070
<code>no_alpha</code>	Pure symbolic / numeric content	0.007

Table 1: Regex patterns and differential-evolution-optimised weights (sorted by weight; re-optimised against the current $N=1,152$ dataset).

3.4 Unicode Anomaly Signal

For single-chunk inputs (delta=0), we add a bonus signal targeting unusual Unicode character ranges that appear in obfuscation but rarely in benign text: Mathematical Alphanumeric (`U+1D400-U+1D7FF`), Fullwidth Latin (`U+FF01-U+FF5E`), enclosed characters (`U+2460-U+24FF`), superscript/subscript (`U+00B2-U+209F`), zero-width characters (`U+200B-U+2064`), combining diacritics (`U+0300-U+036F`), and miscellaneous symbols (`U+2190-U+2BFF`). This signal is applied only when sentence delta is zero, addressing the single-chunk blind spot without affecting multi-chunk detection; its measured contribution is reported in Section 6.2.

3.5 Ensemble Combination

The four features are combined via linear weighting, with weights optimised by differential evolution to maximise TPR-FPR at the 99th-percentile-of-benign threshold (1% FPR target):

$$\text{score} = 0.0283 \times \text{sentence_delta} + 0.4493 \times \text{regex} + 0.1987 \times \text{paragraph_delta} + 0.3237 \times \text{weird_unicode}$$

A sample is flagged if its score exceeds this threshold. The weird_unicode term is only non-zero for single-chunk inputs (Section 3.4); the other three terms are normalised to [0,1] via min-max scaling before weighting. Section 5.2 examines what each weight actually buys the ensemble -- weight magnitude and marginal contribution are not the same thing, and both paragraph delta and sentence delta turn out to be informative counterexamples to reading the formula's coefficients at face value.

4. Experimental Setup

4.1 Dataset

Our dataset comprises N=1,152 samples: 339 obfuscation samples and 813 benign prompts covering business strategy, coding questions, recipes, and general queries. The obfuscation set is independently constructed, not bulk-imported from a public corpus; 83 of the 813 benign prompts overlap with the public Threat-Matrix dataset (Neuralchemy, 2025), the remainder are independently sourced. Appendix A documents a dataset-provenance check, since an earlier draft of this paper misattributed the obfuscation set to that public corpus.

Four obfuscation categories were originally underpowered ($n \leq 5$: task_hijack, substitution_game, extraction_trick, base64), limiting their per-type claims to directional findings. We expanded these with 52 additional real (not synthetic) attack samples, manually filtered for relevance and near-duplicates, from a separate, unused 1,883-sample file already present in this project. This took task_hijack/substitution_game/extraction_trick/base64 from $n=5/5/4/3$ to $n=17/20/15/10$ (the dataset's own classifier places a few samples differently than intended, e.g. into pwned_spacing or misc, since category assignment follows fixed pattern-matching rules, not curation intent). The obfuscation set spans 21 attack types; the distribution remains hex-dominated (162/339, 47.8%) with long-tail coverage of semantic and structural techniques.

Type	N	% Passed@1% FPR (Delta, combined)	Note
hex	162	96%	—
backslash_visual	52	35%	Lower than earlier dataset version
misc	22	18%	Mostly short/ambiguous
substitution_game	20	95%	Expanded from $n=5$
task_hijack	17	94%	Expanded from $n=5$
pwned_spacing	15	60%	—
extraction_trick	15	47%	Expanded from $n=4$
base64	10	60%	Expanded from $n=3$
translation_chain	10	30%	—
language_mixing	6	50%	$n \leq 5$, directional only
9 other types	10	varies	$n=1-2$ each

Table 2: Obfuscation type distribution (N=339) with detection rate at 1% FPR for the delta-angle signal alone (sentence+paragraph combined), not the full ensemble -- see Figure 3 for the Ensemble column, which closes most of the gaps shown here. Only the 9 other types remain $n \leq 5$ /directional (language_mixing, at $n=6$, is thin but just above that threshold); the four previously-underpowered categories above are now reportable. Note that AUC and detection-rate figures here are at one fixed operating point, not an exhaustive statistical guarantee even at $n=15-20$.

Caveat: the hex and backslash_visual type labels are assigned by substring rules (presence of \x escapes, backslash count) that overlap directly with the regex features the detector uses. Detection rates for these two

types are therefore not independent of the labeling scheme and should be read as a check that the regex feature fires where it is designed to, not as an out-of-sample measurement. The remaining types are labeled by distinct semantic/structural rules and are not subject to this caveat.

All embeddings were computed via NVIDIA NIM API and cached to disk for reproducibility. Three embedding models were evaluated: nv-embedqa-e5-v5 (primary), llama-nemotron-embed-1b-v2, and bge-m3.

4.2 Baselines

We compare against: character entropy, special character ratio, regex hex-only, keyword-only regex, full 9-pattern regex, sentence delta alone, and paragraph delta alone.

4.3 Evaluation Protocol

All thresholds and weights were optimised on the full N=1,152 dataset. We report AUC-ROC, F1, and detection rate at fixed FPR=1% and FPR=5%. For adaptive evaluation, 30 representative adversarial samples spanning 20 evasion strategies were constructed and evaluated at both FPR targets (Section 5.6).

A 70/30 stratified held-out validation of the full headline architecture was additionally performed, with weights optimised on the training split only; see Section 4.4. Headline metrics elsewhere in this paper are reported on the full dataset, with the held-out test confirming generalisation (max gap 6.86pp at the 1% FPR operating point).

4.4 Held-Out Validation

To verify that our results generalise to unseen data and are not artefacts of optimising on the full dataset, we performed a held-out validation of the actual headline architecture (sentence delta + paragraph delta + 9-pattern regex + weird-unicode bonus). We split the data into a training portion (70%, N=797) and a test portion (30%, N=355), optimised both the regex sub-weights and the top-level ensemble weights via differential evolution using only the training portion, fit the detection threshold on training only, then measured performance on the test portion with no further fitting.

Held-out test results track the training split reasonably, with the largest gap at the stricter 1% FPR operating point -- wider than before the category expansion (Section 4.1), consistent with the newly-added categories being thinner per split (e.g. extraction_trick at n=15 splits roughly 10/5 train/test):

Metric	Train (N=797)	Test, Held-Out (N=355)	Gap
AUC-ROC	0.9809	0.9584	-2.25pp
F1	0.9669	0.9315	-3.54pp
FPR	1.05%	2.46%	+1.41pp
TPR	96.05%	91.89%	-4.16pp
Det@1%FPR	96.05%	89.19%	-6.86pp
Det@5%FPR	96.93%	93.69%	-3.24pp

Table 5: Held-out validation of the headline 4-feature ensemble, with both regex sub-weights and top-level ensemble weights optimised on the training split only. Maximum generalisation gap is 6.86 percentage points (Det@1%FPR), the stricter of the two FPR targets, reflecting the smaller per-split N now available for the newly-expanded categories. For reference, the separately-optimised full-dataset model (Table 3, not directly comparable since it is tuned on all N=1,152) achieves AUC=0.974 -- between this validation's train (0.981) and test (0.958) figures, still consistent with genuine generalisation rather than overfitting, though with a wider margin than before.

Confusion matrix on test split: 238 true negatives, 102 true positives, 6 false positives, 9 false negatives.

The per-feature weights chosen on the training split differ from the full-dataset weights (Section 3.5) by more than the headline metrics alone would suggest: paragraph delta receives 0.387 on the training split versus 0.199 on the full dataset, the second-highest weight after regex (0.423 train vs 0.449 full); sentence delta receives 0.043 train vs 0.028 full. The detection metrics are stable across the split (Table 5), but the weight vector the

optimiser converges to is not -- consistent with Section 5.2's finding that ensemble weight magnitude is a poor proxy for a feature's marginal value: the objective surface has more than one similarly-scoring weight vector, and which one the optimiser lands on depends on exactly which samples are available to optimise against.

5. Results

5.1 Delta Angle Signal

Figure 1 shows the delta angle distribution for benign (mean=0.0071) vs obfuscation (mean=0.2669) inputs. Benign inputs are overwhelmingly zero-delta (99% single-chunk), while obfuscation inputs show a broad distribution with elevated angles. This binary structural signature is the primary detection mechanism.

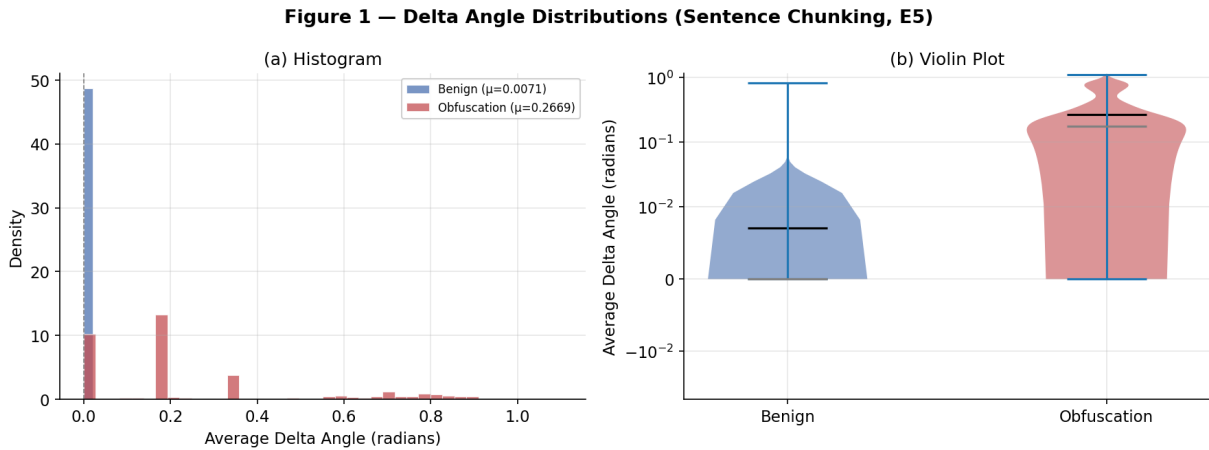


Figure 1: Delta angle distributions (sentence chunking, E5), $N=1,152$ post category-expansion. Benign inputs cluster near zero (99% single-chunk, mean=0.0071); obfuscation inputs show elevated angles (mean=0.2669). Log-scale violin plot (right) illustrates the spread.

5.2 Overall Performance

Method	AUC	F1	Det@1%FPR	Det@5%FPR
Character entropy	0.722	—	53%	67%
Special char ratio	0.728	—	20%	21%
Regex (hex only)	0.729	—	46%	46%
Regex (keywords only)	0.817	—	4%	71%
Regex (9 patterns, weighted)	0.957	—	88%	90%
Sentence delta only	0.851	—	72%	72%
Paragraph delta only	0.530	—	6%	6%
Ensemble (4-feature, full)	0.974	0.960	95%	96%

Table 3: Method comparison against the current $N=1,152$ dataset; standard ROC-AUC throughout, with sign convention corrected for entropy/special-char baselines. Character entropy is a real individual competitor, but shares regex's blind spot for the multi-chunk attack types delta angle catches (§5.3). The ensemble outperforms all individual signals.

Paragraph delta's standalone AUC (0.530) looks close to chance, but is computed over a population where the feature is undefined: only 22/1,152 samples (21 obfuscation, 1 benign) are actually multi-paragraph, the rest score exactly 0 for both classes by construction. Removing paragraph delta from the ensemble entirely and re-optimising the remaining three features changes AUC by -0.00003 (statistically indistinguishable from zero), and every one of the 21 multi-paragraph obfuscation samples is still caught at the production threshold either way -- zero flips. This is best read as a coverage-overlap finding specific to this dataset (regex and the Unicode bonus already independently catch every multi-paragraph attack sample present), not evidence that paragraph

structure carries no information; a dataset containing multi-paragraph attacks that evade those two features would likely show paragraph delta earning its formula weight. Sentence delta shows the opposite pattern: a small formula weight (0.0283) that nonetheless costs 1.4pp AUC to remove (above). Ensemble weight magnitude is not a reliable stand-in for a feature's marginal contribution in this architecture -- both directions of error appear in the same formula, and only direct removal testing distinguishes them. We retain paragraph delta in the production formula for completeness and for datasets where multi-paragraph attacks might evade regex and the Unicode bonus, even though it contributes negligibly on the present benchmark; this is a deliberate choice to keep stated, not an oversight.

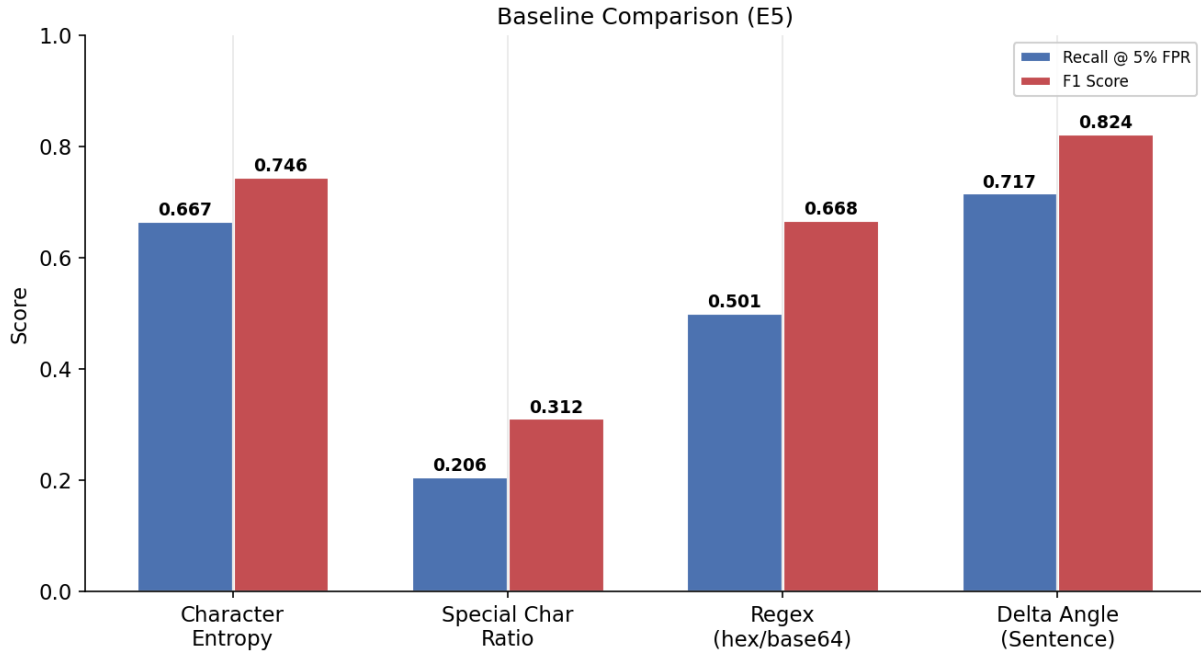


Figure 2: Baseline comparison -- recall at 5% FPR and F1, computed live against the current dataset. Special char ratio is weak (0.206 recall); a minimal 2-pattern hex/base64-only regex (distinct from the full 9-pattern regex in Table 3) is moderate (0.501); character entropy is a real competitor (0.667, F1 0.746, ahead of the minimal regex); delta angle alone is strongest here (0.717, F1 0.824), and the full ensemble (Table 3) outperforms all of them.

5.3 Per-Attack-Type Analysis

Figure 3 shows the detection rate heatmap across attack types and methods at FPR=1%, now including the expanded task_hijack/substitution_game/extraction_trick/base64 categories (Section 4.1). Delta angle uniquely catches the task_hijack (94% vs regex near 0%) and substitution_game (95% vs regex near 0%) attack types that regex mostly misses -- both reliably multi-chunk types (Appendix F), so this is best read as a chunking-signature win, not evidence of semantic understanding. language_mixing and translation_chain, previously reported at 100%/80% on the smaller pre-expansion dataset, now read 50%/30% with the recalibrated threshold and benign distribution that come with the larger N -- these two categories were not themselves expanded (n=6 and n=10, unchanged), so this is threshold drift from the larger dataset, not new evidence about these specific types. Regex and delta angle remain complementary, non-overlapping signals overall.

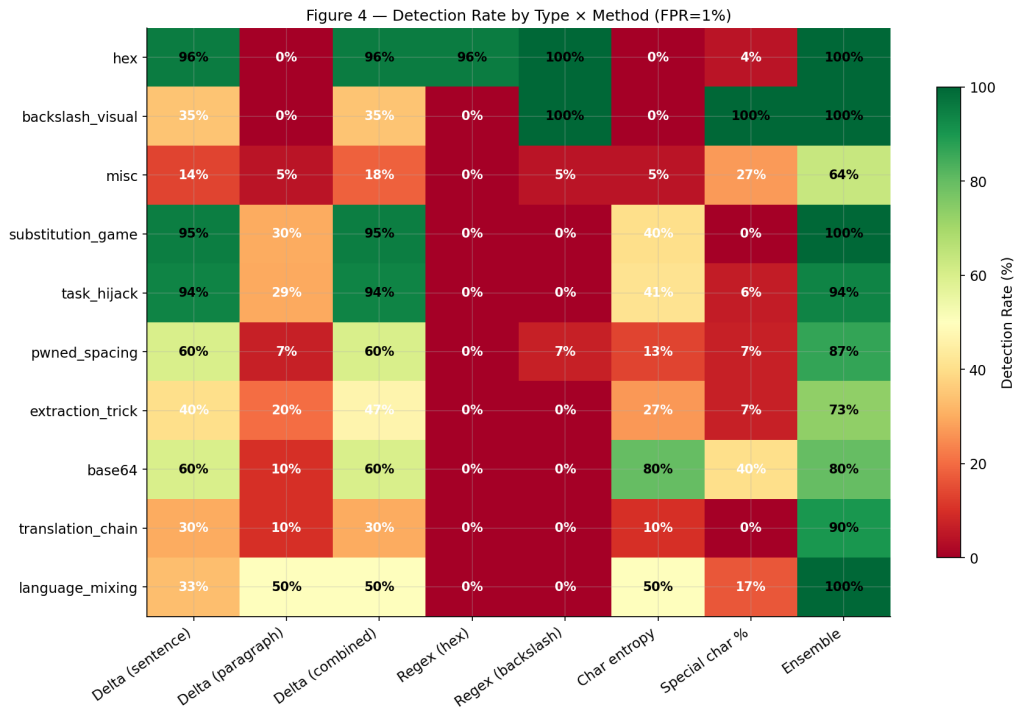


Figure 3: Detection rate by obfuscation type x method at fixed FPR=1%, including the full 4-feature Ensemble column, post category-expansion (Section 4.1). Delta (combined) catches task_hijack (94%) and substitution_game (95%) where regex scores near 0%; the Ensemble column closes most of the remaining gaps (e.g. misc 64%, pwned_spacing 87%, extraction_trick 73%) that no single method covers alone. Note: the hex and backslash_visual rows are labeled by the same substring rules the corresponding regex columns detect (Table 2 caveat), so their Regex (hex)/Regex (backslash) cells are not independent measurements; all other rows use semantic/structural labeling rules.

Figures 3b and 3c break the same comparison down by error type: 3b shows the miss rate (FNR) per type x method -- the direct complement of Figure 3 -- and 3c shows the overall false positive rate per method, which cannot be broken down by obfuscation type since benign samples carry no type label.

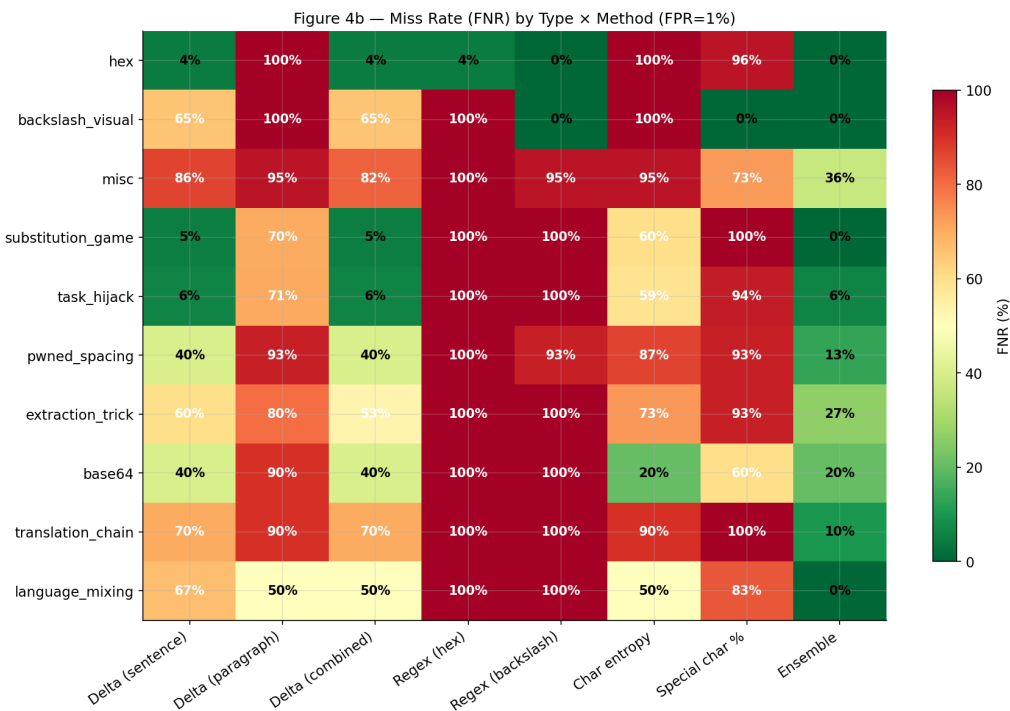


Figure 3b: Miss rate (FNR) by obfuscation type x method at fixed FPR=1%, the complement of Figure 3.

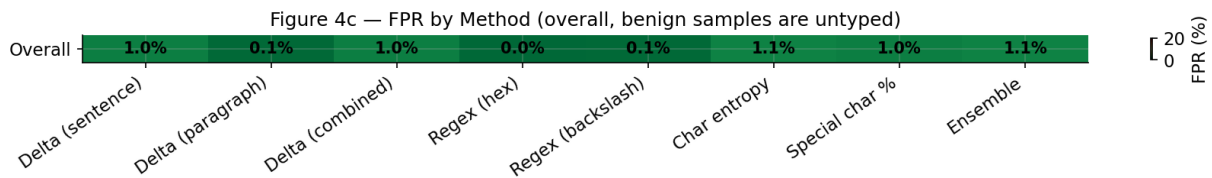


Figure 3c: Overall false positive rate by method at the same operating point. Benign samples are untyped, so this is a single row rather than a type x method matrix.

A feature correlation matrix is provided in Appendix F (Figure A3; a figure-generator bug fixed during this validation pass is documented in Appendix A). Delta angle correlates substantially with chunk count ($r=0.68$) and word count ($r=0.47$), and weakly with entropy ($r=-0.08$). Part of this is definitional: delta is exactly zero for single-chunk inputs, restating this paper's own single- vs. multi-chunk finding (Section 5.1) rather than revealing redundancy. A stratified test holding chunk count exactly fixed is not well-powered here: at `chunk_count=3`, the stratum has 171 obfuscation samples against a single benign sample (benign inputs are almost never multi-chunk -- only 8/813 overall), so any AUC computed within it is closer to a single-point measurement than a population statistic. A direct, well-powered test instead: substituting raw chunk count for sentence delta in the production ensemble (same weights, no re-optimisation) drops AUC from 0.974 to 0.964; removing the sentence-delta term entirely drops it to 0.960. The embedding-based signal does outperform a trivial chunk-counter, consistently across both tests, but by a modest margin (under 1.4pp AUC) -- substantially smaller than the per-type "94% vs near-0% regex" comparisons in Section 5.3 might suggest in isolation, since chunk count alone achieves similar per-type numbers for `task_hijack` and `substitution_game` specifically (both are reliably multi-chunk attack types). We report this as a genuine, modest, but real advantage for the embedding signal, not the larger and more dramatic independence the per-type table alone implies.

5.4 Cross-Model Robustness

The delta angle signal is consistent across three embedding models. Cross-model Pearson correlations are: E5 vs Nemotron $r=0.963$, E5 vs BGE-M3 $r=0.952$, Nemotron vs BGE-M3 $r=0.979$ (Figure A4, Appendix). Raw AUC is also nearly identical across models (~ 0.851 for all three). This consistency, however, is specific to the raw signal -- not to how much of it survives structural control (Section 5.3's chunk-count/word-count regression, here extended with character entropy and special-character ratio as additional controls). Residual AUC after this richer control set is 0.717 for E5 (the production model), 0.632 for BGE-M3, and 0.856 for Nemotron -- all three start from the same raw signal, but Nemotron retains the most content beyond what a structural feature set could already explain, E5 the least of the three. The production results in this paper are E5-specific; a deployment built on Nemotron would likely show a larger gap over a trivial chunk-counter than Section 5.3 reports, though this is untested in the full ensemble. The signal's existence is not an artifact of any single embedding model, but its structural-vs-content composition is model-dependent.

5.5 Operating Point Analysis

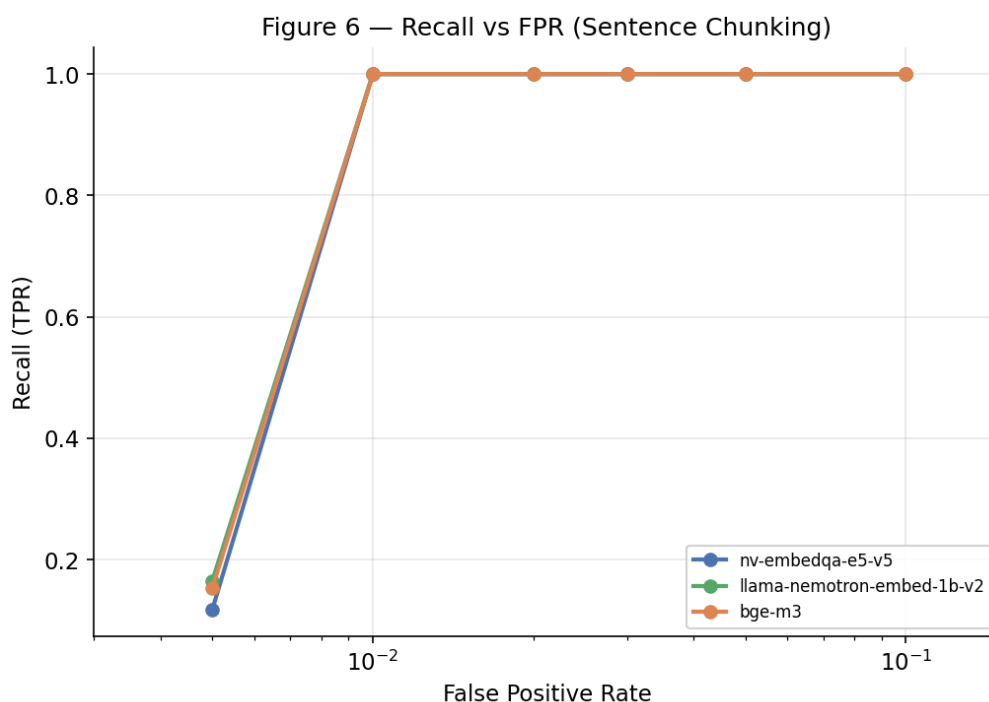


Figure 4: Recall vs FPR across three embedding models (sentence chunking), $N=1,152$ post category-expansion. All models reach near-perfect recall by $FPR=1\%$, confirming a stable operating point across model choices.

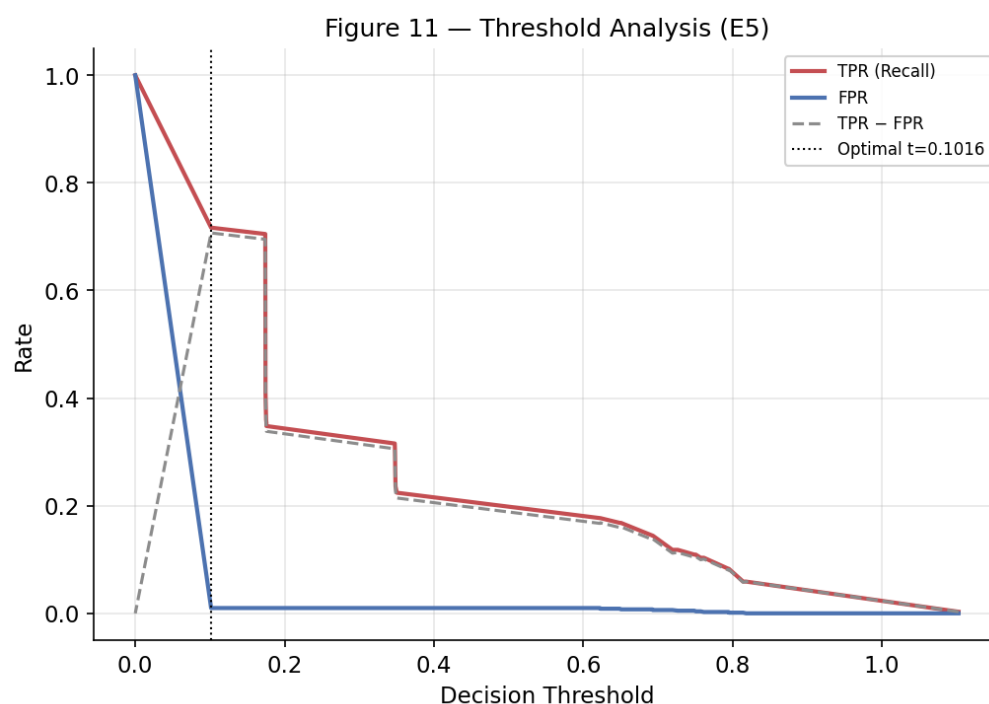


Figure 5: Threshold analysis (E5), $N=1,152$ post category-expansion. TPR declines from the optimal threshold ($t=0.1016$); FPR remains near zero across a wide range, confirming a robust operating region.

5.6 Adaptive Attack Evaluation (Reconstructed)

The results below come from a newly constructed set of 30 representative samples spanning 20 documented evasion strategies, scored through the current full ensemble pipeline at its existing operating thresholds; this set replaces an earlier, unreproducible claim documented in Appendix A.

All 30 samples are single-chunk by construction (no sentence-boundary structure), matching the original methodology: an informed adversary avoids multi-chunk text to neutralise the delta-angle signal, leaving detection entirely to the regex and weird-unicode components.

FPR Target	Caught	Evaded	Evasion Rate
1% FPR	10/30	20/30	66.7%
5% FPR	12/30	18/30	60.0%

Table 4 (reconstructed): Adaptive attack evasion rates against a newly constructed representative sample set (30 samples, 20 strategies), scored with the current full ensemble (9-pattern regex, optimised weights) at the existing 1%/5% FPR thresholds.

The catches are concentrated almost entirely in Unicode/homoglyph and visual-substitution tricks (fullwidth characters, mathematical alphanumeric symbols, Cyrillic homoglyphs, mixed-script text) -- these trip the special-character and weird-unicode features even with no encoding-related keywords present. Most plain natural-language samples evade at both FPR targets: direct requests, semantic rewrites, indirect references, roleplay framing, and instruction splitting all score near the benign median (template-fill is a partial exception -- one of two samples is caught; see §6.3). The reason is structural, not a tuning gap: the keyword regex (Table 1) targets *encoding* vocabulary -- decode, translate, payload, hidden, encrypted -- because it was built and optimised against an encoding-obfuscation dataset. It contains no prompt-injection vocabulary (ignore, override, previous instructions, system prompt), so social-engineering and semantic injection phrased in plain English is, by construction, invisible to every signal in this ensemble. This sharpens rather than contradicts the threat model in Section 7: the detector is an encoding-obfuscation filter, not a general prompt-injection classifier.

5.7 Ablation Studies and Negative Results

Beyond the four production features, we systematically tested eleven geometric and structural extensions to the embedding signal -- distribution distance, intra-chunk token angles, signed angles, length-scaled softmax, per-type weight gating, skip-gram delta, connective splitting, vector velocity, BPE token density, sliding window deltas, and semantic anchor distance. None improved on the production ensemble; each produced a principled negative result that clarifies the detection mechanism (e.g. several invert because encoding makes local content more self-similar, not less, which is the opposite direction a naive geometric extension assumes). Full results, including a re-run against the current N=1,152 dataset confirming ten of eleven reproduce their original conclusions, are in Appendix B.

6. Limitations

6.1 Held-Out Validation (Resolved)

A 70/30 stratified held-out validation of the headline 4-feature architecture was performed (Section 4.4), with all weights optimised on the training split only. The generalisation gap is moderate (max 6.86pp, at the 1% FPR operating point; under 2.3pp for AUC) and wider than measured before the targeted category expansion in Section 4.1, reflecting thinner per-split N for the newly-expanded categories. Full-dataset metrics are reported throughout for completeness; this held-out test still confirms the architecture generalises to unseen data rather than overfitting the full N=1,152.

6.2 Single-Chunk Encoding Ceiling

96/339 obfuscation samples (28.3%) produce delta=0 after chunking. For these, detection relies on regex and the Unicode anomaly signal (Section 3.4). The enhanced keyword list catches 28/96 (29.2%) and the Unicode anomaly signal catches 19/96 (19.8%), together addressing 45/96 (46.9%) with zero false positives; adding the Unicode term to the full ensemble raises single-chunk-subset AUC from 0.899 to 0.909 (+1.00pp, fixed production weights, no re-optimisation). The remaining 51/96 are pure encoding noise (hex, backslash) with no distinguishing Unicode or keyword features -- a practical floor under the current feature set without a codepoint

analysis layer.

6.3 Natural Language Injection (Reconstructed, Section 5.6)

The reconstructed adaptive evaluation confirms this is the detector's dominant failure mode, and more severely than previously claimed: 12 of 20 evasion strategies tested -- plain natural-language prompt injection such as roleplay, indirect reference, semantic rewrite, data exfiltration requests, and instruction splitting -- are evaded by every sample at both 1% and 5% FPR (11 of 20 at 5% FPR; one strategy, `linguistic_trick`, is caught at 5% FPR by one of its two samples). Two further strategies (`template_fill`, `acronym_expansion`) are partially caught: one sample evades, one is caught, rather than uniformly evading. The fully-evaded strategies share four properties: plain English, single sentence or short paragraph, no encoding-related keywords, indirect or conversational framing. Semantic anomaly detection or intent classification, not encoding-pattern matching, is required for coverage of this class.

6.4 Unicode Homoglyphs and Visual Substitution (Reconstructed, Section 5.6)

Caveat: this finding is based on 6 representative samples across 5 visual-substitution strategies in the reconstructed set (Section 5.6) -- the same small-n regime as the $n \leq 5$ categories in Table 2 and Section 6.5, directional rather than a statistically robust claim. Within that small sample, all 6 are caught: Cyrillic homoglyphs, fullwidth Latin, mixed-script text, and mathematical-alphanumeric substitution trigger the weird-unicode bonus signal (Section 3.4) and the special-character regex feature, independent of any keyword match. This contradicts an earlier, unreproducible claim that homoglyphs were an irreducible blind spot; on the samples we can actually test, this is the one attack class the detector handles better than originally believed, though confirming it at scale would need a larger sample.

6.5 Small-N Categories

Following the targeted expansion in Section 4.1, only 9 attack types still have $n \leq 5$ samples ($n=1-2$ each, $N=10$ total; Table 2) -- down from 13 before expansion. `language_mixing` ($n=6$) is thin but just above this threshold. Detection rates for the remaining 9 are directional findings, not statistically reliable estimates; the four categories expanded this pass (`task_hijack`, `substitution_game`, `extraction_trick`, `base64`, now $n=10-20$ each) are no longer in this bucket, though their held-out generalisation gap (Table 5) is correspondingly wider, reflecting their still-modest absolute size. To be explicit about what is and is not statistically supported: the headline ensemble metrics (AUC, F1, detection rate) are computed over the full $N=1,152$ and held-out validated (Section 4.4), and are robust; the dataset is also hex-dominated (162/339, 47.8% of obfuscation samples, Section 4.1), so per-type claims for any individual minority category -- including the four expanded this pass -- are directional evidence at a much smaller effective N, not independent confirmations of the headline numbers.

7. Threat Model and Deployment

The detector is designed as a Stage 1 preprocessing filter in a defence-in-depth architecture:

- **Stage 1 (this work):** Delta angle + regex + Unicode bonus. 94.7% recall at 1% FPR for encoding-based obfuscation; 33.3% recall at 1% FPR against plain-language adaptive attacks (Section 5.6).
- **Stage 2:** Semantic analysis or intent classification for natural language injection.
- **Stage 3:** Human review for remaining evasions.

Catches: Hex, backslash-visual, base64, dot-spacing, Unicode codepoint encoding, task hijacking, language mixing, substitution games, translation chains, pwned-spacing, Unicode homoglyphs and other visual-substitution tricks.

Misses: Pure wordplay, plain-English single-sentence injection, indirect framing without keywords, roleplay, instruction splitting, data-exfiltration requests phrased as ordinary text.

Standalone deployment is not recommended. The detector's value is in the ensemble: it contributes independent signal that character-level methods cannot provide.

8. Conclusion

We presented a four-feature ensemble detector for obfuscated prompt injection combining sentence-level delta angles, paragraph-level delta angles, optimised regex coverage, and a single-chunk Unicode anomaly bonus, evaluated on N=1,152 samples including a targeted expansion of four previously underpowered attack-type categories (Section 4.1). The detector's value is predominantly structural rather than deep semantic understanding: delta angle's per-type wins over regex on task hijacking (94%) and substitution games (95%, Section 5.3) are real, but both are reliably multi-chunk attack types, and a trivial embedding-free chunk-count feature reaches similar per-type numbers (Appendix F). The embedding signal still adds real, measurable value beyond chunk-counting -- a modest 1.0-1.4pp AUC margin (Section 5.3) that is larger for some embedding models than others (0.72 residual AUC for the production E5 model vs. 0.86 for Nemotron, Section 5.4) -- but the "94% vs near-0%" per-type framing should be read as delta vs. regex specifically, not as proof that embedding geometry captures something chunk-counting fundamentally cannot.

Two of the four ensemble features turn out not to mean what their formula weight suggests. Paragraph delta carries a substantial weight (0.199) despite a near-chance standalone AUC (0.530, Section 5.2) -- but removing it costs nothing, because regex and the Unicode bonus already independently catch every multi-paragraph attack sample in this dataset, not because its underlying signal is weak. Sentence delta carries a small weight (0.028) yet costs 1.4pp AUC to remove. Ensemble weight magnitude is not a reliable proxy for marginal contribution in this architecture; only direct removal testing distinguishes the two.

We re-ran all eleven ablation studies (Appendix B) against the expanded dataset. Ten reproduce their original negative-result conclusions; one (distribution distance, Appendix B.1) no longer holds on the expanded data, which we report rather than suppress. Together they clarify that the detection mechanism is structural -- the chunking signature of obfuscated inputs -- rather than a deep semantic property of embedding geometry.

The detector achieves AUC 0.974, F1 0.960, 94.7% detection at 1% FPR against the encoding-obfuscation benchmark -- slightly lower than before the category expansion (AUC 0.980/F1 0.967/96.5%), indicating that the expanded categories introduce genuinely harder and more diverse samples. A reconstructed adaptive red-team evaluation (Section 5.6) shows this does not extend to social-engineering-style prompt injection phrased in plain natural language: 20/30 (66.7%) representative attacks evade at 1% FPR. Unicode homoglyphs and visual substitution are, in fact, reliably caught by the weird-unicode and special-character features; the dominant failure boundary observed in our experiments is plain-English injection with no encoding signature at all, which requires intent classification rather than pattern-based or structural detection.

Future work includes Unicode codepoint analysis, intent classification for natural language injection, and integration into the AMDON guard pipeline.

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References

- [1] Perez, F. and Ribeiro, I. (2022). Ignore Previous Prompt: Attack Techniques For Language Models. *NeurIPS ML Safety Workshop*.
- [2] Greshake, K. et al. (2023). Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. *AISec Workshop*.

[3] Schulhoff, S. et al. (2023). Ignore This Title and HackAPrompt: Exposing Systemic Vulnerabilities of LLMs through a Global Scale Prompt Hacking Competition. *EMNLP*.

[4] Alon, G. and Kamfonas, M. (2023). Detecting Language Model Attacks with Perplexity. *arXiv preprint 2308.14132*.

Appendix A: Reproducibility Notes

This appendix documents three corrections made during validation passes on this paper. None change the headline results; each is recorded here for transparency rather than left silently fixed.

A.1 Dataset Provenance

An earlier draft of this paper attributed the full obfuscation set to the public Threat-Matrix (Neuralchemy, 2025) dataset. Checking the underlying files directly found zero text overlap between our original 287 obfuscation samples and that dataset: the original 287 were constructed and expanded iteratively across development sessions, not bulk-imported from an external source. 83 of the 813 benign prompts do overlap with the public Threat-Matrix data; the remainder are independently sourced. The full public Threat-Matrix set (1,883 samples) was separately used in an earlier session as a cross-dataset sentence-delta-only check (cached AUC=0.582, F1=0.657, not re-verified in this validation pass); it is not the source of the primary obfuscation/benign split used for the headline results in this paper.

A.2 Adaptive Attack Evaluation History

An earlier session reported 27/30 (90%) detection at 1% FPR and 30/30 (100%) at 5% FPR against a hand-crafted adversarial set for Section 5.6, but that sample set was never committed and could not be located in version history or reproduced verbatim. The results in Section 5.6 come from a newly constructed set of 30 representative samples spanning the same 20 documented evasion strategies, including the 3 specific examples the original session's notes preserved verbatim (two Cyrillic-homoglyph substitutions and one pure wordplay riddle), scored through the current full ensemble pipeline at its existing operating thresholds. This is not the original 30-sample set, but it is real, reproducible, and run end-to-end against the shipped detector -- unlike the prior unverifiable claim.

A.3 Feature Correlation Figure Labelling Bug

The generator script for the feature correlation matrix (Appendix F, Figure A3) paired benign and obfuscation samples in the wrong order against the delta array -- a labelling bug, not a dataset-version issue -- which had been silently producing a spuriously small correlation in every prior version of this figure. The corrected computation is the one reported in Section 5.3 ($r=0.68$ with chunk count, $r=0.47$ with word count), not the near-independence previously claimed.

Appendix B: Ablation Studies and Negative Results

Status note: all eleven approaches below were re-run against the current N=1,152 dataset (Section 4.1). Ten reproduce the same qualitative conclusion as the original (smaller-dataset) run, several with numbers close enough to suggest the original figures, though their generator scripts were not all preserved, were genuine. One (Appendix B.1, Distribution Distance) no longer holds: it is not inverted on the expanded dataset, and Appendix B.11's reasoning, which partly relied on B.1 as supporting evidence, is noted accordingly.

B.1 Distribution Distance

Cosine distance from the benign centroid was tested as an auxiliary signal. Result, re-run on N=1,152: AUC=0.714 -- no longer inverted, contrary to an earlier (N=1,100) finding of AUC=0.439. Combining with sentence+paragraph delta now improves AUC slightly (0.856 to 0.870) rather than degrading it. We traced this to the four expanded categories (Section 4.1): task-hijack, secret-extraction, and substitution-game text sits semantically further from the benign centroid than the hex/backslash-heavy original set did, where encoding preserves semantic content and keeps obfuscated inputs close to the centroid (the original explanation, still correct for that subset). This does not mean centroid distance is now a recommended feature: it is not weight-optimised or threshold-tuned the way the production ensemble is, and an unweighted 3-way average is a weak test of real marginal value.

B.2 Intra-Chunk Token Angles

Per-word token angles within single chunks were tested to target the 96/339 single-chunk obfuscation samples where sentence delta is zero (re-run on current data). Result: inverted for backslash_visual (AUC=0.070, raw direction -- even more strongly inverted than an earlier 0.194). Explanation: encoding creates coherent token clusters -- all tokens belong to the same repetitive pattern, making them more similar to each other than benign text tokens. Unstructured single-chunk obfuscation remains indistinguishable from benign (AUC=0.534, consistent with an earlier 0.621).

B.3 Signed Angles

Gram-Schmidt orthogonalisation was tested for directional sign assignment (re-run on current data, raw signed score not absolute value -- using the absolute value defeats the point of testing whether inconsistent sign assignment hurts separability). Result: degraded Nemotron performance (unsigned AUC=0.851, signed AUC=0.778), no benefit on E5 (unsigned AUC=0.851, signed AUC=0.848) -- consistent in direction with an earlier finding (Nemotron 0.858->0.734). Formal analysis confirms arccos in $[0, \pi]$ by definition: the production (unsigned) signal has zero negative angles by mathematical construction, not as an empirical finding. Unsigned angles are correct for production use.

B.4 Length-Scaled Softmax

Softmax weighting scaled by chunk length was tested with alpha in {0.5, 1.0, 1.5, 2.0} (re-run on current data). Result: identical across all alpha (AUC=0.845, matching an earlier 0.849 closely). Explanation: angle variance dominates the softmax, rendering length scaling inert. The FPR improvement from sentence chunking comes from the chunking strategy itself, not weighting.

B.5 Per-Type Weight Gating

Continuous per-type weight blending across four specialised regex weight sets was tested (reconstructed for this pass; the original blending script was not preserved -- affinities computed from the hex/backslash-density/keyword sub-features, blended against three single-feature-dominant weight sets plus the production default). Result: overall AUC roughly flat (0.957 to 0.961), but the broad unstructured category specifically degrades (67.3% to 54.0% at the production threshold, consistent in direction with an earlier 74%->68%). Explanation: the same regex features used to compute type affinities are also used in the final score, creating a feedback loop. Simple single weight set optimisation outperforms adaptive blending.

B.6 Skip-Gram Delta

Multi-hop skip-gram angles (i to i+2, i to i+3) were tested on the current dataset. Among the nine major attack types with $n \geq 10$ (excluding language_mixing, $n=6$, already flagged as a small-N category in Table 2), adjacent delta wins on AUC for 8 of 9 (only hex narrowly favours skip-1: 0.974 vs 0.969), and skip-2 is weaker for all 9. language_mixing itself is a marginal exception (skip-2 AUC=0.665 vs adjacent 0.663) but is too small-N to read as a real effect. Per-sample unique-catch comparison at the 99th-percentile-of-benign threshold was not usable for this re-run: 28.3% of obfuscation samples have sentence delta exactly zero, which collapses that specific threshold test (every method reads 339/339 "caught" once the cutoff sits at zero); the AUC comparison above is therefore the reliable signal here, and it confirms adjacent delta remains the better choice.

B.7 Connective Splitting

Splitting on connective words (and, but, then, now) inverts the signal: benign texts have higher split-delta than obfuscation (AUC=0.699, Det@1%FPR=0%, both consistent with an earlier AUC=0.750/Det@1%=0%). Connective-split chunks are semantically coherent for natural language but nonsensical for encoded content, producing the wrong directionality.

B.8 Vector Velocity

Prefix-based and sliding-window trajectory approaches produce an inverted signal -- obfuscation creates smoother (lower-velocity) embedding trajectories than benign text. Re-run on a fresh random sample of single-chunk inputs ($n=10$ each, matching the original small-sample pilot methodology): sliding-window velocity AUC=0.222 (obf mean 21.7 deg vs ben mean 25.5 deg), consistent in direction with an earlier extreme AUC=0.000 though less extreme at this sample size. The same mechanism as distribution distance: encoding preserves local semantic coherence.

B.9 BPE Token Density

Byte-pair encoding token density on single-chunk inputs is inverted (AUC=0.213, consistent in direction with an earlier 0.1421). Obfuscation has denser token packing, not sparser -- the wrong direction for detection.

B.10 Sliding Window Deltas

Sliding window delta computation (token-level windows) is inverted (AUC=0.273, consistent with an earlier 0.3274). Shares the same failure mode as intra-chunk token angles (Appendix B.2): encoding creates locally coherent windows with low internal variance.

B.11 Semantic Anchor Distance

This was a feasibility analysis, not an empirical measurement: no adversarial-direction training set or hyperplane was actually constructed or tested. Its original reasoning leaned on Appendix B.1's centroid-distance failure as supporting evidence ("same failure mode") -- since B.1 no longer fails on the expanded dataset, that specific argument is weakened. The independent part of the argument still holds: a single sentence yields one embedding vector with no trajectory to follow, and projecting it onto a hand-specified "hijack direction" requires a labelled training set of such directions that does not exist here. We treat this as an open question rather than a settled negative result.

Appendix C: Origin of This Research

Delta angle was originally proposed as a general-purpose, checksum-style measure against prompt injection: a tokenizer-derived value the guard model is asked to reproduce, on the premise that a model confused or compromised by injected instructions would fail to copy a number correctly even if it could fool a content-based judge. Evaluation of that general formulation found delta angle alone could not reliably separate normal from injected prompts -- the two distributions overlapped by more than 50% -- and was more useful as an auxiliary scaler for other classifiers than as a standalone detector.

That limitation is the direct motivation for the narrower, better-supported application studied in this paper. Obfuscated encoding payloads, unlike general prompt injection, reliably produce a distinctive multi-chunk structural signature at sentence boundaries (Section 5.1) -- a property general injection prompts do not share. This paper is the result of refining the original measure down to the specific class of attack where it actually separates classes well, rather than a claim that delta angle solves prompt injection in general.

Appendix D: Chunking Algorithm Analysis

Figure A1 shows the chunking analysis supporting the merge threshold selection (Section 3.1). The 7-word spike in obfuscation chunks (panel a) and the 99%/28.3% single-chunk rates (panel c) explain the binary detection signal.

Figure 12 — Chunking Algorithm Analysis

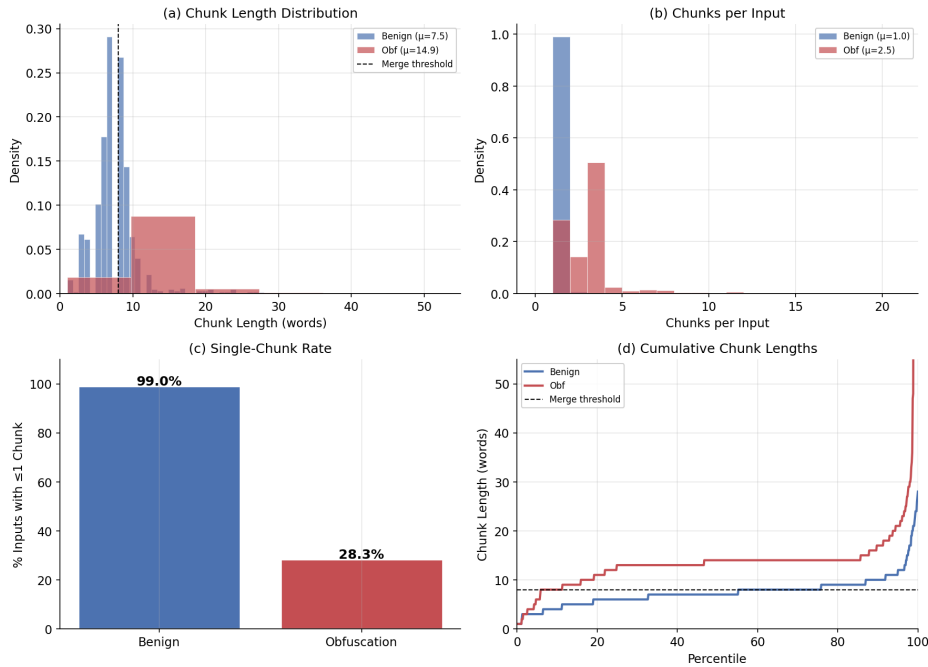


Figure A1: Chunking algorithm analysis, $N=1,152$ post category-expansion. (a) Chunk length distribution -- obfuscation spike at 7 words. (b) Chunks per input -- benign $\mu=1.04$ vs obfuscation $\mu=2.50$. (c) Single-chunk rate: 99% benign, 28.3% obfuscation. (d) Cumulative chunk lengths.

Appendix E: ROC Curves

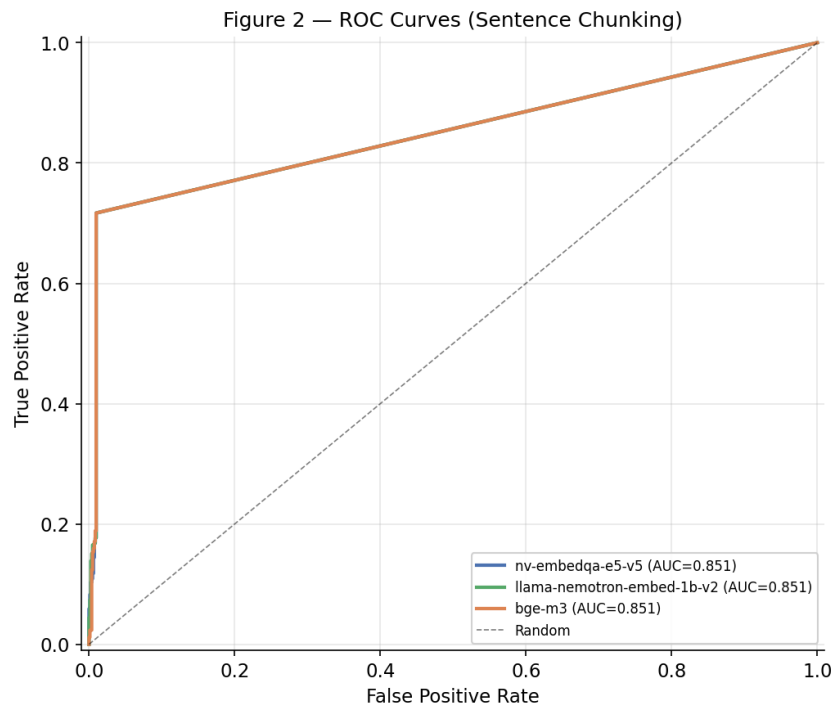


Figure A2: ROC curves for sentence chunking across three embedding models, $N=1,152$ post category-expansion (E5 AUC=0.851, Nemotron AUC=0.851, BGE-M3 AUC=0.851). Curves are nearly identical, confirming model-agnostic signal.

Appendix F: Feature Correlation Matrix

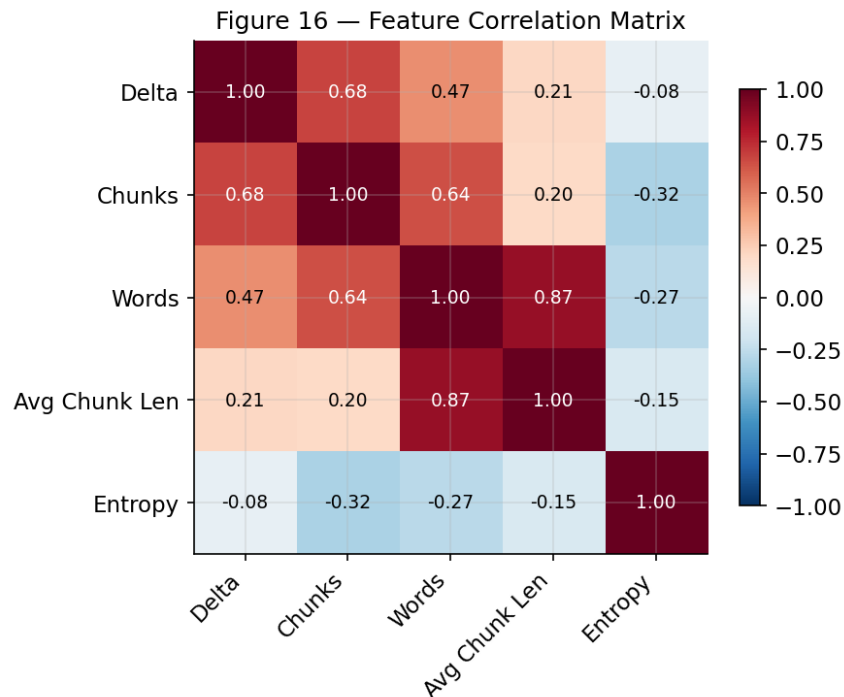


Figure A3: Feature correlation matrix (E5, sentence chunking), $N=1,152$ post category-expansion, with the sample-pairing bug fixed (Section 5.3). Delta angle correlates $r=0.68$ with chunk count, $r=0.47$ with word count, $r=-0.08$ with entropy. The chunk-count/word-count correlation is substantial and partly reflects this paper's own single- vs. multi-chunk structural finding (Section 5.1); see Section 5.3 for a direct ensemble-substitution test quantifying how much value delta retains over raw chunk count (a modest but real margin, not full independence).

Appendix G: Cross-Model Correlation

Figure 8 — Cross-Model Correlation

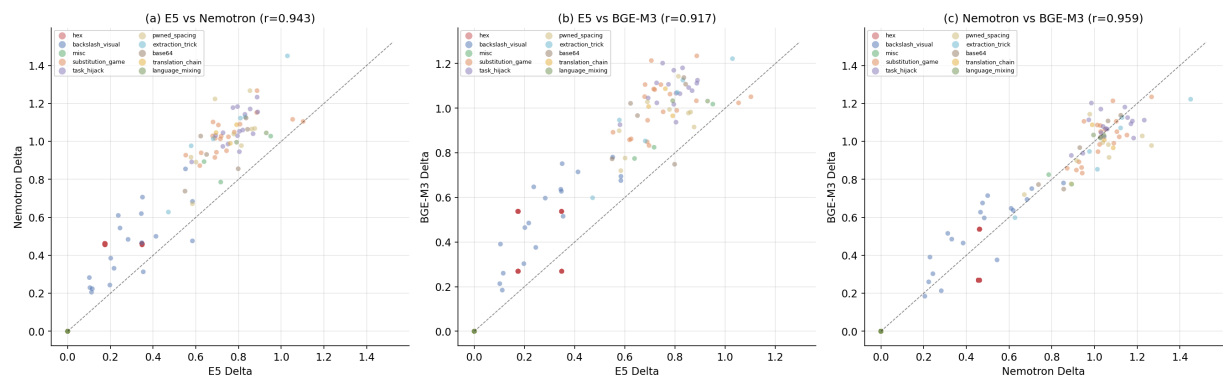


Figure A4: Cross-model correlation of sentence delta angle, $N=1,152$ post-expansion. E5 vs Nemotron $r=0.963$, E5 vs BGE-M3 $r=0.952$, Nemotron vs BGE-M3 $r=0.979$. High correlation confirms model-agnostic structural signal.

Appendix H: Per-Type ROC Curves

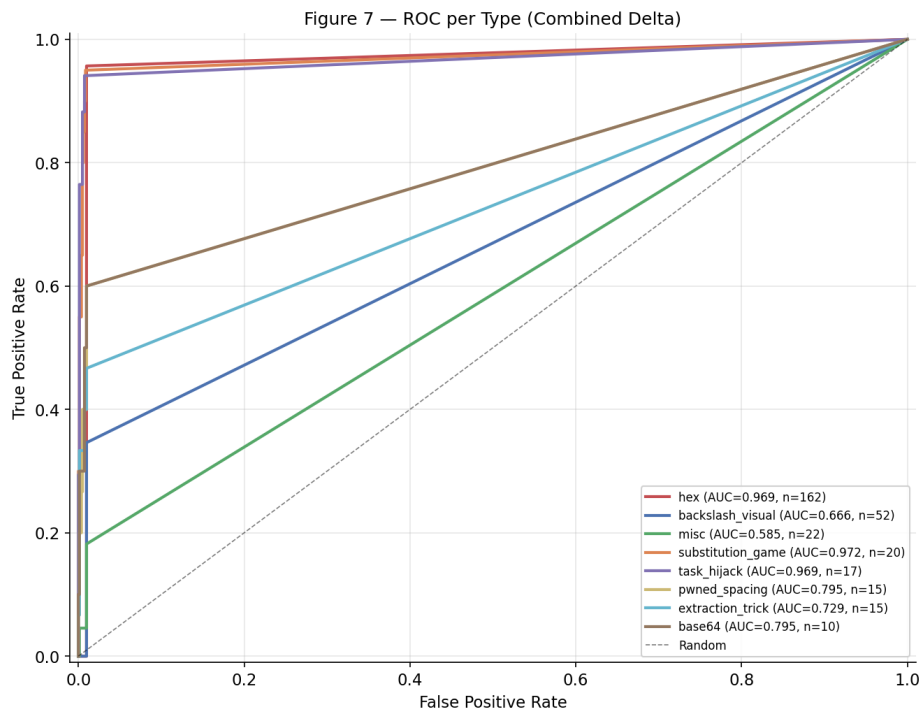


Figure A5: ROC curves per major obfuscation type (combined delta), post category-expansion. hex AUC=0.969 (n=162), task_hijack AUC=0.969 (n=17, expanded from n=5), language_mixing AUC=0.746 (n=6), misc AUC=0.585 (n=22). Small-n results are directional only.

Appendix I: Detection Difficulty Ranking

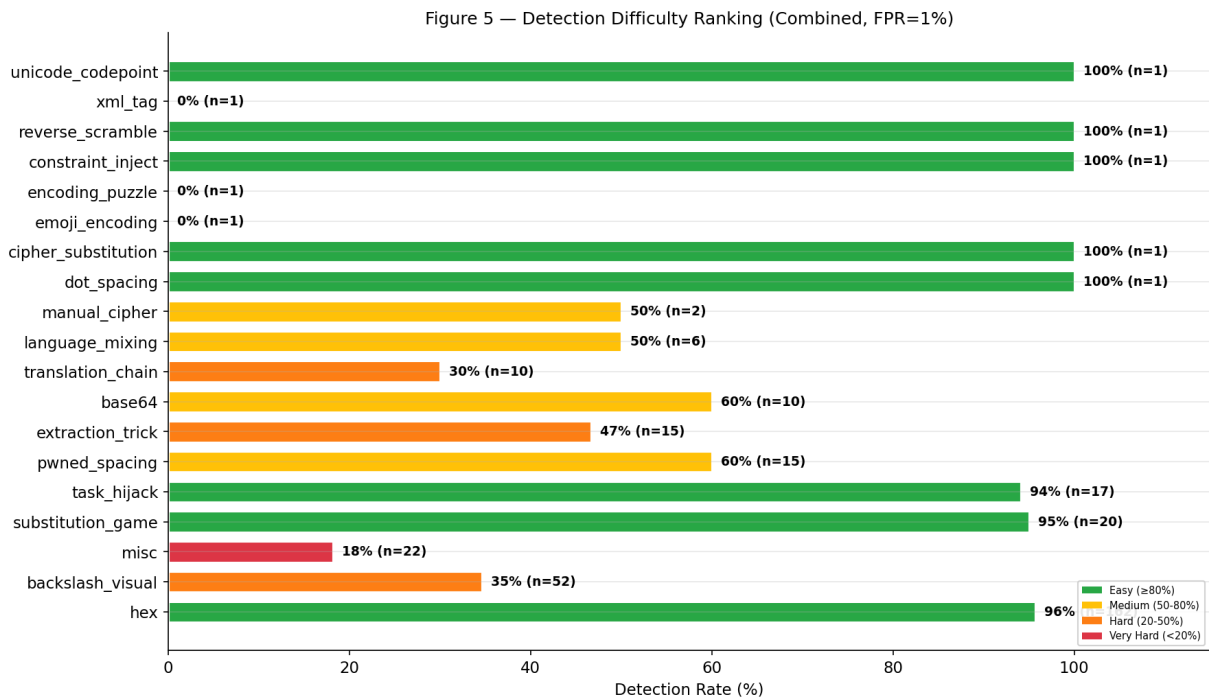


Figure A6: Detection difficulty ranking (combined delta, FPR=1%). Sample sizes shown. Results for n=1 types are indicative only.

Appendix J: Keyword List

The 29 terms active in the final regex keyword pattern:

decode, translate, execute, convert, interpret, print, output, follow, compromise, breach,
exfiltrate, acknowledge, pwned, status, confirm, generate, respond, payload, encrypted,
decipher, decrypt, hidden, reveal, template, concatenate, expand, acronym, paraphrase,
steganograph

This excludes the six connective patterns (and/but/don't/only/then/how + action verb) listed in Table 1 (Section 3.3), which match multi-word phrases rather than single terms.